Full Test — Answers

Final answers with a one-line reason. For full methods, revisit the concept notes.

Objective

Q1	**(b) 360°.** Two triangles × 180°.
Q2	**(b) rhombus.** Diagonals bisecting at right angles.
Q3	**(c) $n(n-3)/2$.** Standard diagonals formula.
Q4	**(b) equal.** Opposite angles of a parallelogram are equal.
Q5	**110.** $360 - (100 + 80 + 70)$.
Q6	**True.** A square has four equal sides, so it is a rhombus.
Q7	**(a) Trapezium.** One pair of parallel sides.
Q8	**(a) Both true, R explains A.** Two triangles give $2 \times 180°$.

Short answer

Q9	**36°, 72°, 108°, 144°.** 10 parts = 360° ⇒ 1 part = 36°.
Q10	**5 cm.** $\sqrt{3^2 + 4^2}$ using half-diagonals.
Q11	**$\angle A = 72°$, $\angle B = 108°$.** Adjacent angles supplementary: $2x + 3x = 180$.
Q12	**$7\sqrt{2} \approx 9.9$ cm.** Diagonal of a square = side × $\sqrt{2}$.

Long answer

Q13	**$\angle C = 70°$, $\angle D = 105°$.** $AB \parallel DC \Rightarrow \angle B + \angle C = 180°$ and $\angle A + \angle D = 180°$.
Q14	**45°, 135°, 45°, 135°.** $x + 3x = 180 \Rightarrow x = 45$; opposite angles equal.
Q15	**(a) 9 sides (b) 140°.** Sides = $360 \div 40$; interior = $180 - 40$.
Q16	**$x = 2$; each diagonal = 5 cm.** Rectangle diagonals equal: $2x + 1 = 3x - 1$.

Total: 28 marks.